



PHASE 1 — FOUNDATION

# Week 1 — Kickoff

Forming the Company

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Say Watt Robotics · ROV Team | 2026-2027 MATE ROV Season

MEETING 1 · THREE HOURS · MADISON, ALABAMA

## THE MEETING

# Three hours, eight stops.

|             |                                                             |               |
|-------------|-------------------------------------------------------------|---------------|
| <b>0:00</b> | <b>Welcome &amp; Safety Brief</b>                           | <b>15 min</b> |
| <b>0:15</b> | <b>What We're Joining — MATE ROV &amp; the RANGER class</b> | <b>30 min</b> |
| <b>0:45</b> | <b>We Are a Company — the six roles</b>                     | <b>45 min</b> |
| <b>1:30</b> | <i>Break</i>                                                | <b>10 min</b> |
| <b>1:40</b> | <b>The Season Ahead — six phases</b>                        | <b>35 min</b> |
| <b>2:15</b> | <b>Building Our Safety Culture</b>                          | <b>20 min</b> |
| <b>2:35</b> | <b>The Community Outreach Log</b>                           | <b>15 min</b> |
| <b>2:50</b> | <b>Wrap-Up &amp; Homework</b>                               | <b>10 min</b> |

## TODAY'S ROLES

# Four jobs. Four students. Different every week.

*Every meeting we assign these four — by season's end, every student has done every job.*

### NOTETAKER / SCRIBE

TODAY: \_\_\_\_\_

Captures every decision, action item, and blocker. Files notes with the CFO for the engineering notebook.

### SAFETY BRIEFER

TODAY: \_\_\_\_\_

Delivers the standing four-question safety brief at the start of the meeting.

### PHOTOGRAPHER

TODAY: \_\_\_\_\_

Captures photos and short videos throughout — feeds the engineering notebook and the marketing display.

### TIMEKEEPER

TODAY: \_\_\_\_\_

Holds the meeting to the agenda's clock. Calls the time at each block change.



SECTION 01

# Welcome & Safety Brief

0:00 - 0:15 · 15 MIN

01

*Every meeting we ever hold will begin in exactly this place.*

# Every meeting opens with a safety brief.

### Why we do this

Real engineering and dive teams brief safety before every job — not because something went wrong, but so it never does. We adopt the same habit on day one. Sixty seconds, every meeting, no exceptions.

Each brief answers the same four questions, so the team always knows what to expect.

- 1 What are we doing today?
- 2 What could hurt someone?
- 3 How do we prevent it?
- 4 Who do we tell if something happens?

### TODAY'S BRIEF · MEETING 1

#### THE WORK

Setup, talking, and planning — no power tools in the room today.

#### THE HAZARDS

Hand tools and box cutters at the build table. The walk to and from the workspace.

#### PREVENTION

Tools stay capped until needed. Closed-toe shoes. Clear the floor of bags and cords.

#### IF SOMETHING HAPPENS

Stop, speak up immediately, and tell the advisor. Small problems get reported too.

#### LOOKING AHEAD

Soldering irons next week and pool sessions soon — the brief grows as the work does.



## SECTION 02

# What We're Joining

0:15 - 0:45 · 30 MIN

*Before we build anything, the team should know exactly what it has signed up for.*

02

## WHAT WE'RE JOINING

# The MATE ROV Competition.

MATE stands for Marine Advanced Technology Education. The MATE ROV Competition is an international contest in which student teams design and build a remotely operated vehicle — an underwater robot — and pilot it through a set of real ocean-industry missions.

The mission tasks are not made up. Each year they mirror real work being done in ocean observation, renewable energy, and keeping waterways healthy. We are practicing the actual job, scaled to a pool.

**Teams advance from a regional contest toward the MATE World Championship.**

### IN ONE SENTENCE

**Design and build an underwater robot — then prove it on real ocean missions.**

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[materovcompetition.org](http://materovcompetition.org)

The official site — manuals, rules, and the mission specs we will live inside all season.

# Our division: the RANGER class.

MATE runs several competition classes. RANGER is the class built for new and intermediate teams — that is us. And RANGER asks for far more than a robot. Five deliverables, one season.

1

## The ROV

The underwater robot itself — piloted on a tether to run the mission tasks.

2

## A profiling float

A second, untethered device that sinks and rises on its own, reporting depth.

3

## Technical documentation

A written engineering report — student-written, decisions explained.

4

## Engineering presentation

A live, spoken presentation to judges. No slides allowed.

5

## Marketing display

A printed display and digital copy that tells the team's story.



## WHAT WE'RE JOINING

# Where this season is headed.

### WHERE

## Dauphin Island Sea Lab

On the Alabama Gulf coast — about a six-hour drive south. The Northern Gulf Coast Regional.

### WHEN

## Late April 2027

Roughly ten months from today. Everything we do is measured back from that weekend.

### WHAT'S AT STAKE

## Advance to Worlds

A strong regional can send the team on to the MATE World Championship.

**Ten months. Around forty-five weekly meetings. We start counting today.**



## SECTION 03

# We Are a Company

0:45 - 1:30 · 45 MIN

*This is the longest block of the day — and the one that decides  
how the team works.*

03

# MATE asks you to run a company.

***“You will not have a teacher running this. You will have a boss — and the boss is each other.”***

MATE deliberately asks teams to operate like a real engineering company, not a class. The reason is simple: it is the fastest way to turn students into engineers. Three things become real the moment we do this.

## **Every job is owned**

No task is 'someone's' job by accident. Each one has a name attached, and that person answers for it.

## **The pressure is real**

Real budget, real deadlines, real reputation. The float either surfaces or it doesn't.

## **You rise together**

A company succeeds as a unit. Your best work means nothing if the team around it fails.

# Six roles. Every student owns one.

## 1 CEO

### CHIEF EXECUTIVE

Owns the schedule, runs the meetings, keeps every sub-team moving toward the deadline.

## 2 CFO

### CHIEF FINANCIAL

Owns the budget and every purchase. Tracks what is spent against what was raised.

## 3 Mechanical Lead

### STRUCTURE & TOOLS

Owns the frame, the float body, and the manipulator — everything that moves or holds.

## 4 Electrical Lead

### POWER & WIRING

Owns the tether, the control box, and every connection that carries current.

## 5 Software Lead

### CODE & VISION

Owns the control code and the computer-vision work that reads the camera feed.

## 6 Communications Lead

### STORY & OUTREACH

Owns the project notebook, the outreach log, and the team's marketing story.

# The roles, up close — leadership & story.

## CEO

Chief Executive Officer

### OWNS

The season schedule, the meeting agenda, and the health of every sub-team. The CEO unblocks people.

### A GOOD FIT IF YOU

like keeping people organized, enjoy seeing the whole picture, and will chase a deadline.

## CFO

Chief Financial Officer

### OWNS

The budget, every purchase order, and the running tally of money spent against money raised.

### A GOOD FIT IF YOU

are careful with numbers, like a tidy spreadsheet, and don't mind asking 'do we really need this?'

## Communications Lead

Story & Outreach

### OWNS

The project notebook, the community-outreach log, and the marketing display and presentation story.

### A GOOD FIT IF YOU

like writing, photography, or talking to people — and want the team's work to be seen.

# The roles, up close — the engineering build.

## Mechanical Lead

Structure & Tools

### OWNS

The ROV frame, the float body, the gripper — every part that holds together, moves, or grabs.

### A GOOD FIT IF YOU

like building physical things, sketching designs, and figuring out how parts fit.

## Electrical Lead

Power & Wiring

### OWNS

The tether, the topside control box, fuses, and every wire termination that carries current.

### A GOOD FIT IF YOU

are patient with detail work, like soldering and wiring, and respect that electricity is serious.

## Software Lead

Code & Vision

### OWNS

The control code that drives the ROV and the computer-vision program that reads the camera.

### A GOOD FIT IF YOU

like programming, want to learn Python and OpenCV, and enjoy debugging until it works.

# How we choose — today, before we leave.

1

## Hear all six

We have just walked every role. Sit with all of them for a moment — no rush.

2

## Write your top two

Each student writes a first and second choice on paper. Two, not one — it keeps us flexible.

3

## Talk it through

We compare picks out loud. Where two students want one seat, we discuss; where a seat is empty, we recruit.

4

## Advisor confirms

The advisor confirms the final set so every role is filled and every student is placed.

**Two things to remember.** No role is lesser than another — and roles can rotate at phase boundaries, so a choice today is not a cage for the season.

B R E A K

# Ten minutes.

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We resume at 1:40 — and, like always, we start on time.





SECTION 04

# The Season Ahead

1:40 - 2:15 · 35 MIN

*Six phases carry the team from an empty room to the regional.  
Here is the whole map.*

04

# Six phases, June to April.

1

## Phase 1 — Foundation

Jun – Jul 2026

WE ARE HERE

**A team that exists, a starter ROV wet, and kids who have soldered.**

2

## Phase 2 — Capability

Aug – mid-Oct 2026

Weekly cadence returns. Each sub-team takes a skill-building mini-project.

3

## Phase 3 — Design

mid-Oct 2026 – mid-Jan 2027

The 2027 manual drops. Decompose missions, run trade studies, lock CAD.

4

## Phase 4 — Build

mid-Jan – Feb 2027

Construction. Frame, electronics, tether, float, and manipulator come together.

5

## Phase 5 — Test & Document

March 2027

Full-mission rehearsals, the JSEA, and the technical documentation.

6

## Phase 6 — Polish & Compete

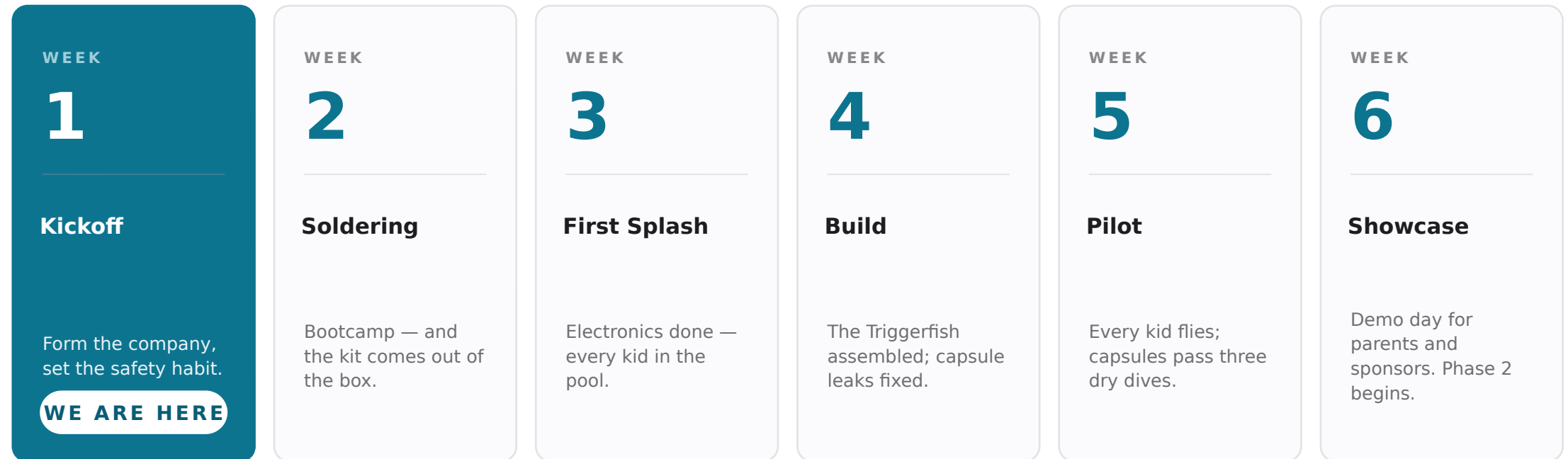
April 2027

Documents in, display printed, presentation rehearsed — travel to Dauphin Island.

## THE SEASON AHEAD

# Phase 1 — our first six weeks.

Phase 1 runs mid-June through July. Six weeks, six themes. We are inside Week 1 right now.



# What Phase 1 must produce.

***“The goal of Phase 1 is not a competition ROV. It is a team that exists.”***

By the end of July, three things are true — and only these three need to be true:

1

## **A team that exists**

Six roles filled, a meeting rhythm that holds, and a safety habit that is automatic.

2

## **A starter ROV in the water**

The SeaMATE Triggerfish built and swimming — every member has piloted it.

3

## **Kids who have soldered**

Real hands-on skill: a clean joint, a sealed capsule, a multimeter read correctly.



SECTION 05

# Building Our Safety Culture

2:15 - 2:35 · 20 MIN

05

*The brief is one minute. The culture is everything that happens  
in the other 179.*

# The habits between the briefs.

A safety brief is a moment. A safety culture is a habit — the way the team behaves when no one is reciting a checklist. Five habits, starting today.

**1 Tool discipline**

Every tool capped, returned, and accounted for. Soldering irons go back in their stands — every time.

**2 The buddy rule**

No one works at the pool alone. Water and a teammate, always — no exceptions, ever.

**3 Respect the current**

Batteries and water do not mix. We treat every electrical step as serious, because it is.

**4 A clear workspace**

Bags off the floor, cords managed, the path walkable. Most injuries are simply a tidy room away.

**5 Speak up early**

Anyone can call a stop. A near-miss gets reported the same as an injury — no one is in trouble for it.

**MATE INSPECTS THIS**

## Safety is judged.

Every ROV passes a safety inspection before it is allowed in the competition water. Loose wiring, sharp edges, or an unsafe team can keep a robot on the deck.

A team that is not safe does not get to compete. We build that habit now, not in April.



SECTION 06

# The Community Outreach Log

2:35 - 2:50 · 15 MIN

*A scored part of the competition that is impossible to fake in April — so we start it on day one.*

06

COMMUNITY OUTREACH

# We start the log on day one.

RANGER teams are recognized for community outreach and engagement — every demo, every younger student taught, every post that shows the work. It counts. And it cannot be reconstructed from memory the week before competition.

**So the rule is simple: if it happened, it gets logged the same week — four columns, nothing fancy.**

| DATE   | WHAT WE DID                       | WHO IT REACHED | REACH |
|--------|-----------------------------------|----------------|-------|
| Week 1 | Team kickoff — company formed     | 6 students     | 6     |
| —      | <i>Your first entry goes here</i> | —              | —     |

WHY DAY ONE

**You cannot remember a year.**

By April, ten months of small moments have blurred together. A log written as you go is honest, complete, and effortless. A log written from memory is neither.

**OWNER** Communications Lead

**OPENS** Today — this meeting is entry one





SECTION 07

# Wrap-Up & Homework

2:50 - 3:00 · 10 MIN

*Close the loop: what each student carries out, and what the team carries forward.*

07

## WRAP-UP & HOMEWORK

# Before we meet again.

### EVERY STUDENT

- 1 Know your role — be able to say it out loud and name what it owns.
- 2 Watch the MATE pre-watch for Class 1.2: Using Tools and Soldering & Waterproofing Wires.
- 3 Bring closed-toe shoes — soldering irons are on the table next meeting.

### THE PARALLEL TRACK · LEADS & ADVISORS

Pool access for the year. MATE registration, waivers, and insurance. The fundraising kickoff. This work runs quietly alongside the build — and it starts now.

### NEXT MEETING · WEEK 2

# Soldering Bootcamp

MATE teaches the concepts as pre-watch homework, so class time goes straight to the bench — every kid solders, and the Triggerfish comes out of the box.

ACTION ITEMS

# What we decided. Who owns it. When it's due.

*The Notetaker fills these in during the meeting and reads them back before we leave.*

| ITEM        | OWNER       | DUE         |
|-------------|-------------|-------------|
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# Tools accounted for. Shop reset.

*We do this together, every meeting. No exceptions — the Tool Steward leads.*

- ☐ Soldering irons unplugged, cool, and put away.
- ☐ All hand tools and test equipment back in their bins.
- ☐ Batteries off charge; charger stations powered down.
- ☐ Workshop swept; benches wiped; trash out.
- ☐ Today's photos uploaded to the team folder.
- ☐ Notetaker's meeting notes filed with the CFO for the engineering notebook.



THE COMPANY EXISTS

# Now we build.

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*By the end of July: a team that exists, a starter ROV in the water, and a room full of students who have soldered something real.*